

|     |                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1 | What is the "fundamental form" of Newton's Second Law of Motion?                                                                                                                                                                     | Force is the rate of change of momentum<br>$F = \frac{\Delta p}{\Delta t}$                                                                                                                                                                                                                                                                                                                                          |
| 2.1 | Is it possible for kinetic energy to increase after a 'collision', where momentum is conserved?                                                                                                                                      | Yes, for example when a gun is fired, or a grenade explodes, or you jump off a boat. Some other form of energy (chemical, gravitational potential, etc.) is being transferred into kinetic energy                                                                                                                                                                                                                   |
| 3.1 | Definition of elastic and inelastic collision.                                                                                                                                                                                       | An elastic collision is one where the total <i>kinetic energy</i> before the collision is the same as the total kinetic energy after.<br>An inelastic collision is one where kinetic energy decreases, lost as it is converted into internal friction (i.e. deformation of object), heat and sound.<br><br><i>most collisions are inelastic, but some "contactless" collisions at the atomic scale are elastic.</i> |
| 4.1 | How do you calculate velocity from momentum?                                                                                                                                                                                         | $p = mv \therefore v = \frac{p}{m}$<br>Divide the momentum by the mass of the object (make sure not to forget that MOMENTUM and VELOCITY are DIFFERENT quantities, even if they are proportional)                                                                                                                                                                                                                   |
| 5.1 | Weighing scales measure weight and not mass (i.e. they require Earth's gravity to make you apply a force $mg$ ). Is there an actual difference? (assuming the scales know the exact acceleration due to gravity you're experiencing) | <b>Yes.</b> Due to the rotation of the Earth, its angular velocity which you also share.                                                                                                                                                                                                                                                                                                                            |

6.1 Your acceleration towards the centre of the Earth (called Earth's gravity) is much stronger than the bare centripetal acceleration required to keep you in your orbit (which has a velocity of  $462 \text{ ms}^{-1}$  at the equator, for example). This difference is felt as the reaction force against the ground. What is the equation relating for the [bare] centripetal force  $F$  in terms of reaction force  $R$  and weight  $W$ ?

$$W - R = F$$

7.1 What else might the 'driving force' of a car be called (especially when viewed from an axis where the car has no velocity, and the car is trying to turn, not drive)?

#### Friction.

The friction of the car's tyres with the road is what allows the car to drive at all— by "opposing the motion of the wheel" the car is forced to move forwards. For a turn, the car doesn't even have to drive; just to angle the wheels *while it has velocity* so that this 'friction with the road' forces the car to turn.

*It makes sense when the only other two forces acting on the car are its weight and the reaction force against the ground.*

*Note that this friction still scales with the velocity of the car.*

8.1 What provides the centripetal force to you while inside a centrifuge? Why does it give the impression of a gravitational field?

Your contact with the ground provides you with the centripetal force (*think: if the centrifuge suddenly disappeared, there would be no force acting to accelerate you towards the centre of some circle, and you would careen off into space*).

And the fact that this contact force is constantly applied (while spinning), and remains constant (in your frame of reference), that makes it similar to the contact force applied to you when it's your weight "pushing" you into the ground (*even though that contact force isn't providing a centripetal force, it's cancelling most of the one given to you by gravity*).

9.1 How do you measure angular velocity (for example in an experiment where your partner is continuously hitting a rubber mass attached to a string)

You measure the time it takes  $t$  to complete a number of revolutions  $N$  using a stopwatch, and use this to calculate angular velocity.

(through  $\omega = \frac{N \times 2\pi}{t}$ )

**NOT:** "the angular velocity of the orbit can be timed with a stopwatch."

$$E_k = \frac{p^2}{2m}$$

10.1 Equation for kinetic energy in terms of momentum, and use case of this equation.

This is useful for subatomic particles travelling non-relativistically (i.e. not near the speed of light), which can have a defined momentum but no defined velocity (which is needed by the conventional  $E_k = \frac{1}{2}mv^2$ ), because velocity is defined as rate of change of position over time, and position may be unknowable thanks to Heisenberg's uncertainty principle.

11.1 What is impulse (definition, unit, equation)?

Impulse is change in momentum; a certain force applied for a certain amount of time (*a lower force applied for a longer time or a higher force applied for a shorter time may GENERATE the SAME impulse*).  
Unit: Ns — kgms<sup>-1</sup>

$$I = F \times \Delta t = \Delta p$$

12.1 What is true about conservation of momentum in multiple dimensions?

Momentum in each dimension is conserved separately.

Investigate The Relationship Between the Force Exerted on an Object and its Change of Momentum.

The objective is to determine the momentum change of a trolley when a force acts on it, as a function of time

**Equipment:**

- Slotted masses and hanger
- Light gate and recorder
- Stop clock
- Metre rule
- Dynamics trolley or air track vehicle
- Bench pulley
- String

**Procedure:**

1. Secure the bench pulley to one end of the runway. The end of the runway should project over the end of a bench, so that the string connecting the mass hanger and the trolley passes over the pulley. The mass hanger will fall to the floor as the trolley moves along the runway. The runway should be tilted to compensate for friction.

2. Place the slotted mass hanger on the floor and move the trolley backwards along the runway until the string becomes tight, with the mass still on the floor. Place the light gate so it is positioned in the middle of the interrupt card on the trolley. There should be enough space on the ramp to allow the trolley to continue so that it clears the light gate before hitting the pulley.

3. Move the trolley further backwards until the mass hanger is touching the pulley. Put slotted masses on the trolley (making sure they will not slide off). This is the start position for the experiment.

4. Record the [...],  $m$ . Release [...], it takes for the trolley [...]— this should be when [...]. Record the time reading,  $t$ , [...]. Repeat

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3. Move the trolley further backwards until the mass hanger is touching the pulley. Put slotted masses on the trolley (making sure they will not slide off). This is the start position for the experiment.
4. Record the **total hanging mass**,  $m$ . Release **the trolley** and use **the stop clock to measure the time,  $T$** , it takes for the trolley **to move from the start position to the light gate**— this should be when **the mass hanger hits the floor**. Record the time reading,  $t$ , **on the light gate**. Repeat **the measurements twice more and calculate mean values for  $T$  and  $t$** . Then estimate  **$\delta T$  and  $\delta t$ , the uncertainties in these values**.
5. Change how many slotted masses are on the trolley vs. on the hanger (keeping **the total combined mass the same**), and repeat the experiment .
6. Measure the combined mass,  $M$ , of the trolley, string, slotted masses and hanger.
7. Measure the distance,  $d$ , travelled by the trolley. This should be the same as the distance fallen by the mass hanger.
8. Record the length,  $L$ , of the card.

#### Safety:

13.5

- [...]. Then estimate [...].
- Change how many slotted masses are on the trolley vs. on the hanger (keeping [...]), and repeat the experiment.
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#### Safety:

- Make sure runways and trolleys (which are very heavy) won't slide or fall off of the bench
- If using an air track, the blower should be on the floor and the hose secured that it won't come loose and blow dust and dirt (potentially into people's faces).
- Large slotted masses need a catch box in the drop zone, cushioned and kept clear of feet or other

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#### Calculation:

The theory we are testing is that impulse = change in momentum:

$$mgT = Mv$$

where  $m$  is the mass on the hanger,  $g$  is Earth's gravity on the surface,  $M$  is the total combined mass, and  $v$  is the speed of the trolley directly after the hanger has fallen to the floor.

To calculate  $v$ , we use:

$$v = \frac{L}{t}$$

where  $L$  is the length of the card (that obstructs the light gate), and  $t$  is the time reading on the light gate.

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Cover Previous  
Reveal Next

Imagine an object moves around part of the circumference of a circle. The angle through which it moves, measured in radians, is defined as the distance it travels, divided by its distance from the centre of the circle.

#### 14.1 Definition of radian

$$\text{angle (in radians)} = \frac{\text{length of arc}}{\text{radius of arc}} = \frac{s}{r}$$

#### 15.1 Definition of angular displacement

The angle through which a body rotates/moves around a circle. Measured in degrees or radians.

#### 16.1 With angular displacement being a vector quantity, what is positive and what is negative angular displacement?

Positive angular displacement is anticlockwise rotation.  
Negative angular displacement is clockwise rotation.

*think of postive displacement as going UP from the positive x-axis.*

#### 17.1 Definition of angular velocity

Change in angular displacement divided by time.  
Symbol:  $\omega$  (omega)  
Unit:  $\text{rad s}^{-1}$

|      |                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18.1 | Equation for angular velocity from period, and equation for angular velocity from frequency | $\omega = \frac{2\pi}{T} \quad \text{where } T \text{ is the period}$ $\omega = 2\pi f \quad \text{where } f \text{ is the frequency}$                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 19.1 | Equation for instantaneous velocity from angular velocity                                   | $v = r\omega$ <p>where <math>r</math> is the radius (in metres) and <math>\omega</math> is the angular velocity (in <math>\text{rad s}^{-1}</math>).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 20.1 | Equation for instantaneous velocity from period or frequency                                | <p>From period:</p> $v = \frac{2\pi r}{T}$ <p>From frequency:</p> $v = 2\pi r f$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 21.1 | Equation for centripetal acceleration from angular velocity or instantaneous velocity.      | centripetal acceleration, $a = r\omega^2 = \frac{v^2}{r}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 22.1 | Equation for centripetal force from angular velocity or instantaneous velocity              | $F = mr\omega^2 = \frac{mv^2}{r}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23.1 | Definition of electric field                                                                | <p>A region [in space] where a charged particle experiences a force</p> <p>OR more broadly, a mathematical way of conceptualising the fact that charges affect each other <i>over space</i>.</p> <p>The force that <i>would</i> be applied to a <u>positive unit charge</u>, if one existed at a point in space, for every point in space.</p> <p><i>The term "electric field strength" would connotate itself as the magnitude of this force, but in fact, it is a vector quantity and IS this force. Some sources use the term synonymously with the term "electric field" itself.</i></p> |
| 24.1 | Definition of electric field strength + unit + symbol                                       | <p>The "electric field strength" is a <u>vector quantity</u> (associated with a given point in space, for a given electric field), denoting the <i>force</i> the electric field would exert on a <u>unit charge</u>, at that point in space.</p> <p><b>Symbol:</b> <math>E</math></p> <p><b>Unit:</b> <math>[\text{NC}^{-1}] / [\text{Vm}^{-1}]</math></p>                                                                                                                                                                                                                                   |

$$F = EQ$$

|      |                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25.1 | Equation for the force exerted on a charged particle, in an electric field        | <p>Where:</p> <ul style="list-style-type: none"> <li>• <math>F</math> is the force exerted on the charged particle</li> <li>• <math>E</math> is the electric field strength of the electric field at the point where the particle is located</li> <li>• <math>Q</math> is the charge of the particle</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 26.1 | What is electric potential? What is the meaning of it being positive or negative? | <p>Electric potential (associated with a given electric field, at a given point in space) is the <i>work</i> the electric field would do on a <u>positive unit charge</u>, at a given point.</p> <p>If the electric field works to push the <i>positive</i> charge "away" from the field (e.g. the field created by a proton), that's positive potential. If you need to work <i>against</i> the electric field to bring the positive charge "away" from the field (e.g. the field created by an electron), that's <i>negative</i> potential.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 27.1 | What is potential difference?                                                     | <p>The difference in <u>electric potential</u> of two (explicitly defined) points in the electric field</p> <hr/> <p>In fact, <i>both "electric potential" and "potential difference" measure the change in electric potential energy (per unit charge) from a <u>reference point</u>.</i></p> <p>It's just that typically, "electric potential" implies that the reference point is <i>infinity</i> (or ground - if that's actually different - or something else standard in general),</p> <hr/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 28.1 | What is the difference between electric potential and potential difference?       | <p>which is to say, it describes the amount of work done against/by the electric field as the charge moves from the given point, to an arbitrary point an infinite distance away (<i>due to the nature of electric fields, the "infinitely far away point" we choose doesn't matter— think that they all exist at the same "hill height", and the path we take to get from one hill height to another doesn't change the difference in gpe between the two; it doesn't change the TOTAL work done by/against gravity to get from one to the other</i>).</p> $V = \int_{\infty}^{\mathbf{r}} \vec{E} \, d\mathbf{r}$ <p>Imagine the 1D case; at each point there's a force applied to the unit charge. "Work" is defined as a force, multiplied by the distance travelled in the direction of that force. Though in this case it's more complicated, as when we travel in the direction of the force, the force itself changes. Hence, it's like the area under a curve with the distance on the x-axis, and the force on the y-axis.</p> <hr/> <p>...while "potential difference" implies that the reference point is another explicitly defined point in our system/circuit.</p> <hr/> |

$$E_p = VQ$$

29.1 Equation for electrical potential energy of a charged particle

Where:

- $E_p$  is the electrical potential energy of the particle
- $V$  is the *electric potential* of the electric field at the particle's location
- $Q$  is the charge of the particle

$$E = -\frac{dV}{dx}$$

30.1 What is the relationship between electric field strength and electric potential?

Electric field strength is the *change in electric potential over distance* (at a given point).  
The force applied to a charge will be 0 if movement doesn't change the electric potential— it's like being on a flat surface; a ball doesn't roll anywhere— i.e. the derivative is 0.  
And the force applied to a charge will be greater if electric potential is changing more rapidly— for example, an electron barreling towards 10 protons instead of 1; the potential will decrease faster over an identical distance since there was more potential to begin with— i.e. the derivative is bigger.

Note: it's the *negative* derivative, because the electric field points from positive to negative potential (it represents the motion of a positive charge), while the conventional gradient will of course point "up" the slope, from negative to positive.

31.1 Difference between electric potential and electric potential energy

**Electric potential** is the work the electric field would do on a *unit charge*.  
**Electric potential energy** is the work the electric field would do on a particle of an *arbitrary charge* (it is a property of the particle, by virtue of its position in an electric field).

32.1 How are electric fields visualized? (2+2)

Using **field lines** (lines with arrows that extend infinitely), which indicate the direction a *positively charged particle* would be pushed.  
And using **equipotentials** (lines which form closed loops, without arrows), which connect all points where the electric potential is equal.  
  
*Neither* field lines nor equipotential lines are allowed to intersect other field lines/equipotential lines.  
The intersection of field lines and equipotential lines are always at *right angles* to each other.

$$k = \frac{1}{4\pi\epsilon_0} = 8.98755 \dots \times 10^9 \text{ Nm}^2\text{C}^{-2}$$

33.1 Coulomb's constant

$$\approx 8.99 \times 10^9 \text{ Nm}^2\text{C}^{-2}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$$

34.1 Permittivity of free space

The "ability of a vacuum to permit electric fields propagating through it"

A measure of "how dense an electric field is permitted to form in response to electric charges"

Relates "the unit of charge to mechanical quantities such as length and force".

Other materials (such as water) may have different "permittivity", as an extra factor  $\epsilon_r$  (relative permittivity) would appear in equations to account for this.

35.1 Equation for the electric field strength at a point distance  $r$  from the centre of a radial electric field around a charge  $Q$

$$E = \frac{kQ}{r^2} = \frac{Q}{4\pi\epsilon_0 r^2}$$

36.1 From a visualization of an electric field, how can you tell where the field is stronger/weaker? (2)

1. The **electric field lines** are packed closer together where the field is *stronger*.
2. The **equipotential lines** are packed closer together where the field is *stronger*.

37.1 Equation for electric potential at a point which is distance  $r$  from a point charge  $Q$

Given the electric field strength  $E = \frac{kQ}{r^2}$  and the electric field strength is the negative maximum rate of change of electric potential  $E = -\frac{dV}{dx}$ ,

$$-\frac{dV}{dr} = \frac{kQ}{r^2}$$

$$\frac{dV}{dr} = -\frac{kQ}{r^2} = -kQr^{-2}$$

$$V = kQr^{-1} \quad \therefore \frac{d}{dr}[kQr^{-1}] = -kQr^{-2}$$

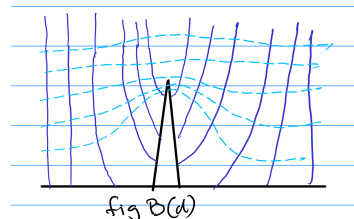
38.1 Why are lightning rods in the shape of a spike? What is the visualisation of the electric field around a spike?

Electrons responding to the electric field will move to the tip of the spike and be *forced close together*, becoming very concentrated.

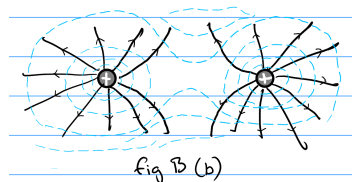
This makes the electric field around the tip very strong,

strongly attracting the opposite charge (the lightning), and hence reducing the chance for the lightning to strike elsewhere.

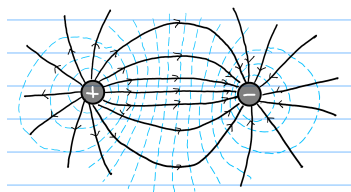
The visualisation:



39.1 Visualisation of electric field around two positive charges spaced apart



40.1 Visualisation of the electric field around two particles of opposite charges spaced apart.



41.1 Coulomb's law equation

$$F = \frac{kQ_1Q_2}{r^2} = \frac{Q_1Q_2}{4\pi\epsilon_0r^2}$$

42.1 What does the sign of the result of the Coulomb's law equation represent?

When the force calculated from Coulomb's law is *negative*, that means the particles are attracted towards each other.  
...and if the calculated force is positive, then they are repelled away from each other.

43.1 Expression for force exerted on charged particle by other charged particle of identical charge,  $Q$ , a distance  $r$  away.

$$F = \frac{kQ^2}{r^2}$$

44.1 Avg. diameter of atoms  
And hence, the average cross-sectional area of an atom

avg. diameter of atoms  $\approx 1.2 \times 10^{-10} \text{ m}$

avg. cross-sectional area is therefore  $\approx 1.1 \times 10^{-20} \text{ m}^2$

45.1 Most common atom/ number of electrons we can assume in estimating contact forces.

**Oxygen**, which has **8 electrons**.

Hence, 4 electrons on the contact side, per atom

46.1 What is a capacitor?

A capacitor is an electric circuit component  
...that stores charge (and so can be used as an *energy* store)

47.1 What is capacitance + symbol + unit

Capacitance, **symbol**:  $C$ , **unit**: F (farads) [ $\text{CV}^{-1}$ ]  
A measure of the *capability* of a capacitor  
(a property of the capacitor object/component)

$$C = \frac{Q}{V} \quad \therefore Q = CV$$

Where:

- 48.1 The defining equation for capacitance
- $C$  is the capacitance of the capacitor
  - $Q$  is the charge stored by the capacitor
  - $V$  is the potential difference across the capacitor

$$\text{capacitance (F)} = \frac{\text{charge stored (C)}}{\text{potential difference across capacitor (V)}}$$

- 49.1 Equation for energy stored by capacitor in terms of charge stored  $Q$  and p.d. across capacitor  $V$

$$E = \frac{1}{2}QV$$

- 50.1 Equation for energy stored by capacitor in terms of its capacitance  $C$ , and p.d. across the capacitor  $V$

$$E = \frac{1}{2}CV^2$$

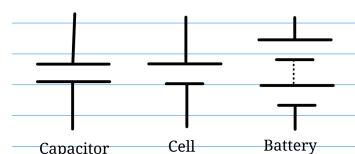
- 51.1 Equation for the energy stored by a capacitor, in terms of the capacitance  $C$ , and the charge stored by the capacitor  $Q$

$$E = \frac{1}{2} \cdot \frac{Q^2}{C}$$

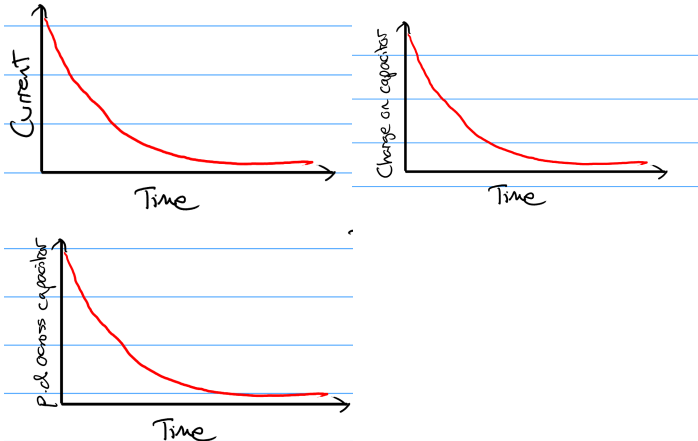
- 52.1 When you double the charge on a capacitor, what happens to the energy stored by it?
- Doubling the charge on a capacitor **quadruples** the energy stored by it.
- In order to add every extra bit of charge across the capacitor, it has to be done by creating a larger p.d. across the capacitor (overcoming the force of all the like-charges already present on the plates), which by definition is the energy transferred per unit charge. Hence, every extra bit of charge added has more energy associated with it, and this grows quadratically*

- 53.1 What is the relation between the potential difference across a capacitor and the charge stored by the capacitor?
- They are **proportional**.
- A plot of charge stored on the y-axis and potential difference on the x-axis will be a straight line passing through the origin with gradient  $C$  (capacitance of the capacitor).  
i.e. the capacitance of a capacitor doesn't change based upon the p.d./charge on it.

- 54.1 Electrical symbols for capacitor, cell, and battery



**Note:** A battery is just multiple cells connected in series.

|      |                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 55.1 | <p>When the two ends of a <b>charged capacitor</b> are connected to form a closed loop (perhaps through a resistance), what is shape of the curve of current through this closed loop over time? And charge on the capacitor over time? And p.d. across the capacitor over time? And why?</p> | <p>The amount of charge on the capacitor dictates the strength of the electric field, which is the p.d. When allowed to discharge, the charge moves according to the strength of the electric field. Thus, the current is high at first, but as the charge leaves (at a rapid rate— because high current) the electric field <i>weakens</i> rapidly, causing current to decrease rapidly. When the current is smaller, the rate at which charge leaves the capacitor is slower, hence the electric field weakens at a slower rate, hence the current weakens at a slower rate, etc. This is <b>exponential decay</b>.</p>  |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|      |                                                  |                                                                                                                                                                                                        |
|------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 56.1 | <p>What is the time constant + symbol + unit</p> | <p>Time constant, <b>symbol:</b> <math>\tau</math> (tau), <b>unit:</b> s (seconds).<br/>The time constant provides an <u>indication</u> of the amount of time that charging/discharging will take.</p> |
|------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

$$\tau = RC$$

|      |                                       |                                                                                                                                                                                                                                                                                             |
|------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 57.1 | <p>Equation for the time constant</p> | <p>Where:</p> <ul style="list-style-type: none"> <li>• <math>\tau</math> is the time constant</li> <li>• <math>R</math> is the resistance of the closed loop across which the capacitor is discharging</li> <li>• <math>C</math> is the capacitance of the discharging capacitor</li> </ul> |
|------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|      |                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 58.1 | <p>What will cause the amount of time for capacitor charge/discharge to occur to lengthen?</p> | <ul style="list-style-type: none"> <li>• If the capacitor is placed in a circuit with more <i>resistance</i>, the current will be smaller (<math>I = V/R</math>) and time charging/discharging will be longer.</li> <li>• If the capacitance of the capacitor itself is greater, the amount of charge stored at the same voltage/current will be larger, and hence the time charging/discharging will be longer.</li> </ul> |
|------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

$$Q = Q_0 e^{-\frac{t}{RC}}$$

*Derivation:*

The p.d. across the capacitor,  $V_c = \frac{Q}{C}$ .

The p.d. across the resistor,  $V_R = IR$ .

Given we're taking the capacitor p.d. as an e.m.f. while discharging,

the total e.m.f. in the circuit is  $-V_c$ , and

the total p.d. in the circuit is  $V_R$ .

$$\begin{aligned} -V_c &= V_R \\ -\frac{Q}{C} &= IR \quad \therefore -IR = \frac{Q}{C} \end{aligned}$$

The current through the resistor is the rate of change of charge over time in the capacitor:

$$I = \frac{dQ}{dt}$$

That means,

$$\begin{aligned} -R \cdot \frac{dQ}{dt} &= \frac{Q}{C} \\ \frac{dQ}{dt} &= -\frac{Q}{RC} \end{aligned}$$

Essentially, we can see that we're saying "the rate of change of  $Q$  is proportional to  $Q$ " which is what we intuitively understanding, thinking about it physically.

Now though, we have to find some expression for  $Q$  which satisfies this equation (requiring us to solve a differential equation).

$$\frac{1}{Q} \cdot dQ = -\frac{1}{RC} \cdot dt$$

We can integrate this equation, from the start time  $t = 0$  to  $t$  (the independent variable),

...which is equivalently from the initial charge in the capacitor  $Q_0$ , to the current charge  $Q$  (the dependent variable).

$$\begin{aligned} \int_{Q_0}^Q \frac{1}{Q} \cdot dQ &= \int_0^t -\frac{1}{RC} \cdot dt \\ \ln Q - \ln Q_0 &= -\frac{t}{RC} \\ e^{\ln Q} &= e^{\ln Q_0 - \frac{t}{RC}} \\ Q &= e^{\ln Q_0} e^{-\frac{t}{RC}} \\ Q &= Q_0 e^{-\frac{t}{RC}} \end{aligned}$$

To show we are correct, we can substitute this equation back in to verify:

$$\begin{aligned} \frac{dQ}{dt} &= -\frac{Q}{RC} \\ \frac{d}{dt} [Q_0 e^{-\frac{t}{RC}}] &= -\frac{1}{RC} \cdot [Q_0 e^{-\frac{t}{RC}}] \\ Q_0 \left( -\frac{1}{RC} \right) e^{-\frac{t}{RC}} &= -\frac{1}{RC} (Q_0) e^{-\frac{t}{RC}} \\ &\text{...is correct} \end{aligned}$$

59.1 Equation for charge on capacitor  $Q$  after time spent discharging  $t$

|      |                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 60.1 | Equation for p.d. across the capacitor, a time $t$ after it began discharging in an R-C circuit, AND<br>Equation for current in R-C circuit a time $t$ after beginning charging/discharging. | $V = V_0 e^{-\frac{t}{RC}}$ $I = I_0 e^{-\frac{t}{RC}}$                                                                                                                                                                                                                                                                                                                                                               |
| 61.1 | <i>One time constant</i> after a capacitor began discharging in an R-C circuit, what percentage of the original charge is left on the capacitor?                                             | 37%<br><br><i>Or mathematically, <math>e^{-1}</math></i>                                                                                                                                                                                                                                                                                                                                                              |
| 62.1 | After 5 <i>time constants</i> , what percentage of the original current will still be flowing through the R-C circuit?                                                                       | 0.1%<br><i>Mathematically, <math>e^{-5} = 0.0067 \dots</math></i>                                                                                                                                                                                                                                                                                                                                                     |
| 63.1 | What is magnetic flux + unit + symbol                                                                                                                                                        | Magnetic flux, <b>symbol:</b> $\Phi$ (phi), <b>unit:</b> Wb (weber).<br>The "amount" of magnetic field passing through an area (oriented in 3D space)                                                                                                                                                                                                                                                                 |
| 64.1 | What is magnetic flux density + symbol + unit                                                                                                                                                | Magnetic flux density, <b>symbol:</b> $B$ , <b>unit:</b> T (tesla)<br>The strength of the magnetic field, at a given point, measured as ...the <i>force</i> the field applied ( $N$ ), per unit current ( $A^{-1}$ ), per unit length of the conductor oriented perpendicularly to the field ( $m^{-1}$ )<br><i>coming from the identity of magnetic fields as the relativistic effect applied to moving charges.</i> |
| 65.1 | If charges are what create/ get affected by the <i>electric</i> field, what analagous thing is there for the magnetic field?                                                                 | [Magnetic] <b>poles</b> .<br>Specifically, North poles and South poles, <u>which always exist in pairs</u> .                                                                                                                                                                                                                                                                                                          |
| 66.1 | How are magnetic fields visualised? (4)                                                                                                                                                      | Using <b>magnetic field lines</b> , which are lines that<br>...form closed loops<br>...that don't intersect with other field lines<br>...that have arrows on them, that<br><br>indicate the direction a <u>north pole</u> would be pushed                                                                                                                                                                             |

|      |                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 67.1 | Three different ways objects may relate themselves to the magnetic field | <ol style="list-style-type: none"> <li>1. Some objects set up a <u>long-lasting</u> magnetic field around themselves*. For example, <b>neodymium permanent magnets</b>.</li> <li>2. Other objects become magnetised (setting up their own magnetic field) <i>in response</i> to an <u>external magnetic field</u>. For example, <b>iron</b>.</li> <li>3. Other objects are electrically conductive, and hence become magnetised (setting up their own magnetic field) in response to a <u>changing external magnetic field</u> (<i>either because the object moves through the field or the field itself changes</i>). For example, <b>copper</b>.</li> </ol> <p>* <i>Some more unnecessary notes on permanent magnets:</i></p> <ul style="list-style-type: none"> <li>- <i>electrons/protons are tiny magnets (a fundamental property of particles on the quantum level, like charge or mass) i.e. they have intrinsic magnetism. (the fact the electrons are whizzing around does also create magnetic fields, but these are a separate phenomenon and generally cancel out)</i></li> <li>- <i>atoms with half-filled shells let the intrinsic magnetic fields of the electrons add up without cancelling out -&gt; such as iron, cobalt, nickel, chromium (these exist in clusters on the periodic table)</i></li> <li>- <i>atoms aligning in crystals can either do so with all the poles lined up (ferromagnetic), or cancelling each other out (anti-ferromagnetic)</i></li> <li>- <i>these crystals are grouped up to form everyday objects; each one a "domain" inside that object. An external magnet can cause all the domains to line up and magnetize the object (in the case of magnetizable materials), or they can all be lined up to begin with (a permanent magnet).</i></li> </ul> |
|------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

$$\phi = B \times A \sin \theta$$

Where:

|      |                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 68.1 | Equation for magnetic flux passing through an area | <ul style="list-style-type: none"> <li>• <math>\phi</math> is the amount of magnetic flux through the area</li> <li>• <math>B</math> is [some average of] the magnetic flux density (magnetic field strength) at the position of the area</li> <li>• <math>A</math> is the area</li> <li>• <math>A \sin \theta</math> is the component of the area perpendicular to the magnetic field lines (<i>in 3D, this would be the dot product between the <math>\vec{B}</math> vector and the <math>\vec{A}</math> vector (the area vector; with direction perpendicular to the area surface, and magnitude the area size)</i>)</li> </ul> |
|------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The area through which the magnetic flux on a coiled wire is considered is the area through which the coil *encloses* (not the cross-sectional area of the wire itself or anything like that). Assuming it's coiled in a circle, the flux on a single "turn" of wire is:

$$\phi = B \times A \sin \theta \quad \text{where } A = \pi r^2$$

... and the flux on the TOTAL coil is:

$$\text{flux linkage} = N\Phi$$

where  $N$  is the number of turns of the coil.

This is called "flux linkage"; "the amount of magnetic flux interacting with a coil of wire".

Flux linkage, **symbol**: None, **unit**: weber-turns

*If the idea that magnetic flux acts through the space the wire encloses feels strange, then here are some more things to note:*

69.1 What is the magnetic flux acting on a coil of wire? What is this quantity called? What is its unit?

- The concept of "magnetic flux" to begin with is only usefully defined in order to define/apply Faraday's law, which describes how charges (probably electrons) in a wire can be made to form a current in a circuit— a closed loop of wire— by a magnetic field. This is how the idea of thinking about the closed loops formed by wires becomes relevant.

- You can take the understanding of Faraday's law one step further by considering how magnetism is just a relativistic effect of the electric force. You look at how electrons in the circuit move as the overall circuit is made to take different motions (e.g. translated in a uniform magnetic field, rotated in different ways in a uniform magnetic field, translated into a uniform magnetic field from outside, falling downwards while oriented horizontally through a conductive metal tube), and then see from the electrons' frame of reference how the charges (stationary protons and moving electrons) which are creating the magnetic field, change in the apparent speed at which they move, and hence the amount they contract by according to special relativity, which changes the charge density, that may create a resultant electric force on the electrons. (The key tool for translating to/from this understanding is that electrons flowing clockwise through a loop will create a North pole on top, and a South pole on the bottom). Working through these examples you gain an understanding that a current really is only formed in cases where this idea of ""magnetic flux"" acting on the circuit changes.

Faraday's law only really applies when we're considering loops electrons are taking as a whole (and even then you should be aware that the "law" comes from a qualitative understanding, not a physical one. And that in fact, there are cases where the law even apparently "breaks down", creating paradoxes).

It is used for determining the motion of electric motors.

Specifically, the direction of the force applied to a section of wire considering the direction of current in the wire, and the direction of the magnetic field through the wire.

- **Index finger**: points in the direction of the field
- **Middle finger**: points in the direction of the current
- **Thumb**: Indicates the direction of motion/ the direction of the applied force.

70.1 What is Fleming's left-hand rule?

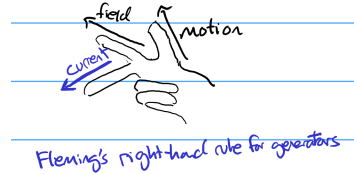


Fleming's right-hand rule is used for determining the direction of current of electric generators.

Specifically, the direction of current induced in a section wire, considering the the direction of motion of the section of wire, and the direction of the magnetic field through the wire.

- **Thumb:** points in the direction of motion of the wire
- **Index finger:** points in the direction of the magnetic field
- **Middle finger:** Indicates the direction of the current induced.

71.1 What is Fleming's right-hand rule?



72.1 What is the motor effect?

The **motor effect**: A wire carrying current will experience a *force* from a magnetic field, if held within one.

73.1 Current travels clockwise through a loop of wire while inside a uniform magnetic field with its North pole on the left and its South pole on the right. What force, if any, will be applied to the loop of wire?

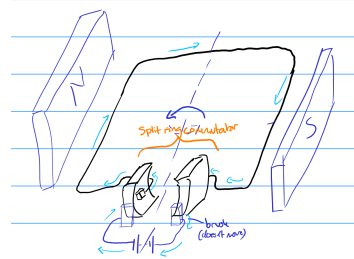
A **clockwise moment** will be applied to the loop of wire.

74.1 What is an electric motor?

An electric motor is made from a coil of wire surrounded by permanent magnets (or electromagnets)

By using a **split ring commutator**, which ensures the current flowing through each side of the loop inside the magnetic field is always constant, by having each side of the loop end in some contact surface that rotates with it, coming into contact with a "brush" at each side which connects to the cell.

75.1 How does an electric motor constantly keep spinning in the same direction? What happens if we don't do this?



If we didn't do this (and for example had the cell rotate together with the loop), then after half a rotation the moment induced on the loop by the magnetic field would be reversed, since the direction the current flows reverses.

$$F = BI \times L \sin \theta$$

Where:

76.1 Equation for the [magnitude of the] force applied to a current-carrying wire in a magnetic field

- $B$  is the magnetic flux density
- $I$  is the current flowing through the wire
- $L$  is the length of wire in the magnetic field
  - $L \sin \theta$  is the component of length perpendicular to the magnetic field

*When studying a situation with the wire perpendicular to the magnetic field, we get the simplified, affectionately named, "Bill equation";*

$$F = BIL$$

77.1 How can one make an electric motor stronger? (3)

- Increase  $I$ , the current in the wire
- Increase  $L$ , the length of wire in the field (e.g. more turns on the coil)
- Increase  $B$ , the magnetic field strength (e.g. more current running through electromagnets)

$$F = Bq \times v \sin \theta$$

Where:

78.1 Equation for the [magnitude of the] force applied to a charged particle in a magnetic field

- $B$  is the magnetic flux density
- $q$  is the charge of the particle
- $v$  is the velocity of the particle
  - $v \sin \theta$  is the component of velocity perpendicular to the magnetic field

*When the particle in question is an electron (charge 'e') moving perpendicular to the magnetic field, we get the simplified, and affectionately named, "Bev equation";*

$$F = Bev$$

79.1 How do charged particles move through magnetic fields?

The direction of the force applied is always **perpendicular** to the velocity (and also the magnetic field, but that's not as relevant), hence it acts as a centripetal force. Charged particles end up making orbits in magnetic fields

80.1 What is Faraday's law?

**Faraday's Law:** Induced e.m.f. on a circuit is proportional to the rate of change *over time* of total magnetic flux through a circuit  
Specifically,

$$\mathcal{E} = -\frac{d\Phi}{dt}$$

When calculating e.m.f. induced across a coil of wire, remember that each turn of the coil adds to the total magnetic flux going through the "circuit".

81.1 What is Lenz's law?

**Lenz's Law:** When a magnetic field induces an e.m.f. across a circuit, the direction of induced e.m.f. is such as to *oppose* the change creating it.  
In other words, the magnetic field created by the induced current will counter the magnetic field that induced said current.

*It is the "-" sign in the Faraday's Law equation.*

|      |                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 82.1 | When a magnet (such as a neodymium permanent magnet) is dropped through a pipe made of conductive material (such as copper), its speed is smaller than normal. Explain this phenomenon. | <p>When the magnet falls down the copper pipe, each section of the pipe experiences change in magnetic flux through it, which by Faraday's Law causes an e.m.f./current in the pipe. By Lenz's Law, the current generated— which will have their own magnetic fields— must oppose the magnetic field that induced them. Hence, the magnet faces opposing magnetic fields as it falls and thus falls slower.</p> <p><i>You can also understand why it would be necessary for the magnet to fall slower through energy conservation— there is electrical energy as current is being induced in the pipe, which MUST come from somewhere, hence it comes from the kinetic energy of the falling magnet.</i></p> <p><i>This experiment is uniquely challenging to interpret from a special relativity understanding.</i> First, we consider the magnet to be a loop of wire carrying current. Let's say the electrons flow clockwise, and hence a South pole is created on the bottom.</p> <p>Consider a view of this falling ring from the perspective of electrons in the pipe. Sections of the ring further away from the point of view of the electron appear at a higher point in space (the further you look, the further back in time you observe), since the ring is falling. Electrons flowing in the falling ring that are moving towards our reference electron will thus also appear to be moving downwards slightly (since they are moving from some higher point in space to some lower point), and vice-versa for electrons flowing away.</p> <p>This changes the amount of vertical contraction— electrons flowing towards are vertically contracted <i>more</i> while electrons flowing away are vertically contracted <i>less</i>. More vertical contraction means a greater proportion of the charge is distributed <i>horizontally</i>, and so a smaller amount of charge that our reference electron sees. This ends up leaving net negative charge on the side that has electrons moving away, and a net positive charge on the side where they flow towards our reference electron. Hence, our reference electron will be compelled to move towards the side where electrons move towards it i.e. in the opposite direction as the electrons in the falling loop. This creates the correct result of now effectively having two magnets with south poles facing each other, slowing the fall of our falling loop.</p> |
| 83.1 | When the magnetic flux on a loop of wire <i>increases</i> , what is the sign of the induced e.m.f.?                                                                                     | <p>The e.m.f. will be <i>negative</i>, due to Lenz's Law</p> <p>A positive gradient on a graph of flux vs. time will result in a negative e.m.f. since <math>\mathcal{E} = -d\Phi/dt</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 84.1 | How can the capacitance of a capacitor be increased? (1+1/1)                                                                                                                            | <ul style="list-style-type: none"> <li>• Make the area of the plates bigger</li> <li>• Reduce the distance between the plates, done in conjunction with...</li> </ul> <p>Improve the quality of the insulator separating the two plates</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 85.1 | What is the experiment that informed the modern understanding of the structure of atoms?                                                                                                | <p>The alpha particle scattering experiment.</p> <p>...which involved aiming <b>alpha particles</b> (<i>two protons and two neutrons, 0 orbiting electrons, i.e. a helium-4 atom nucleus</i>) at a <b>thin gold foil</b> in a <b>vacuum</b>, and then subsequently detecting the alpha particle no matter the angle it is deflected by the gold foil (e.g. a <i>zinc sulfide screen</i> surrounding the gold foil and a <i>microscope</i>).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|      |                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |                                                                                                                                  | <p><b>Thomson's plum pudding model.</b> (1906)</p> <p>That is, a model where atoms consisted of <i>negatively charged electrons</i>, within a <i>positively charged "dough"</i>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 86.1 | Before Rutherford's nuclear model of the atom, what was the prevalent atomic model?                                              | <p>This came from the discovery of electrons— small, negatively charged "bits" that could be added and removed from atoms.</p> <p>...aaand this discovery of electrons came from studying cathode rays (electrons streams)</p> <ul style="list-style-type: none"> <li>- finding that they could travel much farther through air than expected of something atom sized (as suggested by the theory of them being hydrogen atoms)</li> <li>- calculated mass by "measuring heat generated when the ray hit a thermal junction (-&gt; <i>kinetic energy(?)</i>) and comparing that with the magnetic deflection of the rays (-&gt; <i>mass/charge ratio(?)</i>)."</li> </ul> |
| 87.1 | What is Rutherford's nuclear model of the atom?                                                                                  | That there is a large amount of empty space, through which electrons orbit through, around a dense [mass], positive[ly charged] "nucleus"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 88.1 | How do the results of Rutherford's alpha particle scattering experiment inform the Rutherford nuclear model of the atom? (2+2+3) | <p>The <i>majority</i> of alpha particles are deflected relatively little -&gt; Most of the atom is empty space (no/little mass or charge)</p> <p><i>Some</i> alpha particles are deflected a large angle -&gt; There is a large concentration of charge at a point</p> <p>A <i>few</i> alpha particles are deflected more than 90°, reflecting off the gold foil -&gt; By conservation of momentum, the alpha particle must be colliding with something more massive than it -&gt; Most of the mass is densely concentrated in a small, central point.</p>                                                                                                               |
| 89.1 | Mass of neutrons compared to protons                                                                                             | Neutrons are 1% more massive than protons.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 90.1 | How was it recognised that a particle like the neutron was needed in atomic nuclei?                                              | <p>Once the <i>existence of protons</i> was recognised (and their mass and charge)—<i>probably by comparing different elements of the periodic table, measuring mass and seeing how many electrons they accept</i>— it was also recognised that nuclei were <b>too massive</b> considering the number of protons they should contain.</p> <p>This led to the discovery of neutrons (Chadwick, 1930).</p>                                                                                                                                                                                                                                                                  |
| 91.1 | What could the role of neutrons in the nucleus be said to be?                                                                    | Neutrons give physical separation between the protons, which repel each other through the electromagnetic force, binding them together through the strong nuclear force.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 92.1 | How does Quantum mechanics change the model of the atom?                                                                         | <p>According to Quantum mechanics, we cannot simultaneously know the exact position and velocity of a given particle (the Heisenberg uncertainty principle)</p> <p>-&gt; This means electrons do not "orbit" the nucleus in predictable paths (like solar system orbits), but instead exist in '<u>probability clouds</u>', referred to as 'electron <u>orbitals</u>'.</p> <p><i>There's this video about this that may be interesting (I keep ketting recommended it):</i><br/> <a href="https://www.youtube.com/watch?v=6tZXS1dL5A">https://www.youtube.com/watch?v=6tZXS1dL5A</a></p>                                                                                  |
| 93.1 | 'Nucleons' definition                                                                                                            | Particles—protons and/or neutrons—within an atomic nucleus.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 94.1 | 'Atomic number' definition (3)                                                                                                   | <p>A property of an atom;</p> <p>the number of protons in the nucleus of the atom,</p> <p>determining what element (of the periodic table) an atom will be</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|       |                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 95.1  | 'Mass number' definition                                                                   | A property of an atom;<br>the total number of <i>nucleons</i> comprising the nucleus of the atom.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 96.1  | Symbols for atomic number and mass number                                                  | Atomic number: <b>Z</b><br>Mass number: <b>A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 97.1  | 'Isotopes' definition                                                                      | Atoms of the same element (i.e. having the same number of protons in their nuclei), but differing in the number of neutrons.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 98.1  | Explain the following notation: ${}^{226}_{88}\text{Ra}$<br>Give the way you read it aloud | "Radium-226"<br>The superscript indicated the mass number (A) (the <i>specific "flavour" of the element we're describing; encoding the neutron number</i> )<br>The subscript indicates the atomic number (Z) (the <i>element we're talking about, gives the same info as the lettering "Ra"</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 99.1  | What is 'thermionic emission'?                                                             | The emission of electrons from the surface of a metal through the <b>heating</b> of the metal—giving the electrons enough <i>thermal energy</i> to escape.<br><br><i>This poses it as superficially similar to the photoelectric effect, though quite distinct.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 100.1 | What is a 'cathode ray'?                                                                   | A beam of electrons                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 101.1 | Applications of electron beam technology (4)                                               | <ul style="list-style-type: none"> <li>• Electron beam crystallography</li> <li>• Electron microscopy</li> <li>• Cathode-ray oscilloscopes (CROs)</li> <li>• Cathode-ray tube TVs (CRTs)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 102.1 | Three different types of particle accelerator (+1 bonus)                                   | <ol style="list-style-type: none"> <li>1. Linear accelerators (LINACs)</li> <li>2. Cyclotrons</li> <li>3. Synchotron</li> <li>4. +Cyclosynchotron (<i>Cyclotron that takes into account relativistic effects</i>)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 103.1 | What is a linear accelerator, and how does it work?                                        | <p>A linear particle accelerator is a machine which accelerates <i>charged</i> particles along a <i>straight line</i>.</p> <p>A <b>single-stage linear accelerator</b> is the simplest particle accelerator; particles are accelerated through a static (<i>unchanging</i>) p.d. through a tube. (<i>This approach is limited by the maximum single p.d. we are able to achieve; which could be up to 30-100MeV</i>)</p> <p>A <b>multi-stage linear accelerator</b> involves a series of charged tubes(? electrodes?), each of an opposite electric potential, all hooked up to an <i>AC voltage supply</i> at a <u><i>fixed frequency</i></u>. The <u><i>lengths of the pipe vary</i></u> (later stages being longer) so that charged particles spend an <i>equal amount of time</i> through each, accounting for the acceleration (increasing velocity) of the particle.</p> |
| 104.1 | What happens when particles approach relativistic speeds?                                  | <p><i>Relativistic speeds -&gt; Speeds approach the speed of light</i></p> <p>The particle's [apparent(?)] <b>mass increases</b>.</p> <p>...so that their <i>kinetic energy</i> can grow unbounded (in its same simple rule), even while its speed never exceeds (or equals) the speed of light.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**Cyclotron:** A [circular] machine that accelerates charged particles along a spiral path.

Particles take semicircular paths while on the inside of either of two *oppositely charged 'dees'*. (*The electric field inside a hollow conductor is always 0*). This is achieved using a uniform magnetic field that the dees are placed within.

When particles *exits a dee and enter the other one*, they are under the effect of an *electric field that accelerates them*.

Each time the particle velocity increases, the *radius of its orbit increases*, hence as the particle accelerates it spirals outwards, until eventually it goes through a *hole in the side of one of the dees* to collide with something.

What is a cyclotron  
105.1 and how does it  
work? (1+3+3+2)

The frequency at which the dees switch polarity is constant (-> ~~they are hooked up to a constant AC voltage~~... NOT; see below), which *works* because the frequency required for successful acceleration *doesn't depend on radius*.

...for speeds that AREN'T RELATIVISTIC.

(where the relativistic mass isn't significantly changing)

It is timed(?) such that the polarity of the dees switches *immediately* when particles exit a dee; the voltage follows a square-wave pattern (which isn't AC voltage, but a more specialised voltage supply).

*This may be because the electric potential of the dees is not allowed to change while the particle moves inside them; in order to keep the electric field within them 0 (as that is a fact said to be taken from electrostatics ([StackExchange post](#))), or not generating a spurious magnetic field that can mess with the particle paths, or something like that.*

$$f = \frac{Bq}{2\pi m}$$

Where:

- $f$  is
- $B$  is the strength of the [uniform] magnetic field that the dees are within
- $q$  is the charge of the particle being accelerated
- $m$  is the mass of the particle being accelerated

Equation for the  
frequency a cyclotron  
106.1 must switch polarity  
at.  
Includes a derivation.

Derivation:

$$f = \frac{1}{T} \quad \Rightarrow \quad T = \frac{2\pi r}{v}$$

(frequency is related to period and period is related to circumference and the [instantaneous] velocity of the particle)

$$F = Bqv \quad \text{and} \quad F = \frac{mv^2}{r}$$

$Bqv = \frac{mv^2}{r}$  (Relates the strength of the magnetic field and the radius of the orbit it will cause a charged particle to travel in).

$$Bq = \frac{mv}{r} \quad \Rightarrow \quad \frac{r}{v} = \frac{m}{Bq} \quad \Rightarrow \quad \frac{2\pi r}{v} = \frac{2\pi m}{Bq}$$

(recognise the LHS as an expression for the period,  $T$ )

$$\Rightarrow \quad T = \frac{2\pi m}{Bq}$$

$$\Rightarrow \quad f = \frac{Bq}{2\pi m}$$

|       |                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                                                                                   | <p>A synchrotron is a machine that accelerates particles around a <i>fixed circular path</i>.<br/>...As opposed to cyclotrons, that accelerate particles in an outwardly spiralling path.</p> <p>In a synchrotron, particles are accelerated at <i>various points</i> around the loop<br/>...whereas in a cyclotron, they are (TYPICALLY) accelerated at only <i>two points</i> of every orbit<br/>(for cyclotrons with two dees— cyclotrons with more dees also exist)</p>                                                                                   |
| 107.1 | <p>What is a synchrotron?</p> <p>What is the difference from a cyclotron? (4)</p> | <p>In a synchrotron, the voltage frequency varies, in order to keep particles on the fixed circular path, and is <i>able to account for relativistic effects</i><br/>...in a cyclotron, the voltage frequency is <i>constant</i>, meaning it's <u>unable</u> to account for relativistic effects.</p> <p>In a synchrotron, the magnetic field strength <i>varies</i>, in order to keep particles on the fixed circular path (strength increasing as particle velocity increases)<br/>...whereas in a cyclotron, the magnetic field stays <i>constant</i>.</p> |
| 108.1 | How are synchrotrons able to achieve higher energies than cyclotrons? (2)         | <p>By adjusting both the voltage frequency and magnetic field strength, the synchrotron is able to keep accelerating particles <i>even when relativistic effects become large</i>.</p> <p>Keeping particles on the <u>same path</u>, allows <i>more compact</i> and hence higher strength magnets than in cyclotrons.</p>                                                                                                                                                                                                                                     |
| 109.1 | What is the principle of (almost) all particle detectors?                         | <p><b>Ionisation.</b></p> <p>Utilising the charge of the particle to detect and observe it.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 110.1 | Basic operation of a Geiger-Müller detector                                       | <p>The caught particle passes into a tube <i>filled with gas</i>— ionising the gas,<br/>...producing a pulse of electricity<br/>...that is then digitally counted.</p>                                                                                                                                                                                                                                                                                                                                                                                        |
| 111.1 | What is a bubble chamber?                                                         | <p>A vessel filled with <i>superheated, transparent</i> liquid.</p> <p>Superheated -&gt; The liquid is heated <i>beyond its boiling point</i>, <u>without boiling</u>. (Boiling means changing from a liquid into a gas).</p> <p><i>Apparently, this phenomenon can be quite explosive, as the cause is surface tension preventing small bubbles from expanding, but once a bubble has sufficiently expanded to overcome the tension force it only gets easier to expand, which is an evidently explosive process.</i></p>                                    |
| 112.1 | What makes tracks inside a bubble chamber visible?                                | <p>The charge of the particles moving through the chamber, ionising the liquid to form expanding bubbles, that eventually become visible (<i>as they start out microscopic</i>).</p>                                                                                                                                                                                                                                                                                                                                                                          |
| 113.1 | Why do tracks in a bubble chamber spiral?                                         | <p>Due to a uniform magnetic field that is (typically) applied across the bubble chamber, giving a centripetal force to any moving charged particles<br/>...causing the particles to take orbiting trajectories<br/>...and as the particles <i>lose energy through interacting with the liquid</i>, they slow down and hence <i>spiral inward</i>.</p>                                                                                                                                                                                                        |

$$r = \frac{mv}{Bq} = \frac{p}{Bq}$$

Where:

- $r$  is the radius of the orbit the particle will move along
- $m$  is the mass of the particle
- $v$  is the (instantaneous) velocity of the particle
- $q$  is the charge of the particle
- $B$  is the magnetic field strength

Equation + Derivation  
for the radius of the  
114.1 orbiting path of a  
charged particle in a  
magnetic field.

Derivation:

$$F = Bqv \quad \text{and} \quad F = \frac{mv^2}{r}$$

$$Bqv = \frac{mv^2}{r}$$

$$Bq = \frac{mv}{r}$$

$$r = \frac{mv}{Bq} = \frac{p}{Bq}$$

Whether it spirals inward in a *clockwise or anticlockwise* direction tells you the *charge* of the particle (whether it's *positive or negative*)  
(you also have to know the direction of the magnetic field, from which you can then apply Fleming's left-hand rule, where the direction of the "current" is the direction the charged particle travels—or opposite the direction, if it has negative charge—and the induced motion/force is the centripetal force, hence pointing inwards on the spiral).

The size of the spiral can be used to determine the relative mass/charge ratios. "More massive particles have larger radii." -> Charge does not vary so freely as mass (generally, integer multiples of the elementary charge), so is not so important to consider.

---

$r = \frac{mv}{Bq}$  We can calculate radius  $r$  (distance from the centre of the spiral) at any point along the track, and we know the magnetic field strength  $B$ .

$$Br = \frac{mv}{q} = R(\text{igidity})$$

This illustrates that indeed, more massive particles will have a larger orbit (i.e. bigger spiral). It also scales the influence of the velocity  $v$  on the orbit, so we expect the spiral to *decay faster*.

(at this point though, we may still confuse a massive, slow-moving particle with a small, fast-moving particle, which is why we also need...)

In fact, to *calculate* the particle mass and charge, we also measure the bubble density, which will be *proportional* to the change in particle energy over distance (each bubble represents a "fixed minimum energy desposition into the liquid"). We calibrate the chamber with some known particle, and then we can use imaging of the track to find bubble density, leading to loss of particle [kinetic] energy over distance. Using the Bethe-Bloch formula relates this quantity to the particle velocity and charge.

By experimenting with different charge values (multiples of the elementary charge), and perhaps using other clues, this system of two equations and three unknowns can be solved. (the relation between "energy loss over distance", charge and especially speed, is nicely intuitive. Slower speed -> more time spent through the same distance -> more energy loss, AND Higher charge -> more ionisation events -> more energy loss)

What do the  
properites of a visible  
spiral in a bubble  
115.1 chamber tell you  
about the particle that  
made it?

$$\Delta E = c^2 \times \Delta m$$

(this is Einstein's famous equation—  $E = mc^2$ )

Where:

- $\Delta E$  is an amount of energy (J)
- $c$  is the speed of light ( $3.0 \times 10^8 \text{ms}^{-1}$ )
- $\Delta m$  is an amount of mass (kg)

116.1 Relation between energy and mass

*It's a bit hard to understand exactly why this is true. It is a fact that pops out of special relativity, especially when you accept the existence of massless particles (like photons) carrying energy/momentum— you need to preserve momentum conservation in moving reference frames for events like a body at rest emitting two pulses of light in opposite directions (in the lab frame, they have the same frequency, and of course both travel at the speed of light— momentum is naturally conserved; in a moving frame, they both still move at light speed (because this is special relativity, that's the whole point) but the frequencies shift due to the Doppler Effect changing/imbalancing their energy/momentum. The body appears to also move in the moving frame, but its velocity can't be what what changes to 'fix'/balance the momentum conservation, so it must be its mass—the body loses mass!) (in the lab frame case, the body still loses some mass, but it doesn't matter for momentum conservation— the body isn't moving, it doesn't have momentum).*

117.1 What is pair production?

A particle and its anti-matter equivalent are created simultaneously, from a conversion of energy.

118.1 What is annihilation?

The phenomenon in which a particle and its anti-matter equivalent meet, and destroy each other simultaneously  
...from a conversion of mass -> energy, which is carried away by *force-carrier particles*, such as photons.

119.1 Convert  $1 \text{ MeV}/c^2$  to kg

$$\begin{aligned} 1 \text{ MeV}/c^2 &= \frac{1 \times 10^6 \times 1.6 \times 10^{-19}}{(3 \times 10^8)^2} \\ &= 1.78 \times 10^{-30} \left[ \frac{\text{kg m}^2 \text{s}^{-2}}{\text{m}^2 \text{s}^{-2}} \right] = [ \text{kg} ] \end{aligned}$$

(no need to remember the final number  $1.78 \times 10^{-30}$ , just the formula for conversion).

$$\begin{aligned} 1 \text{ kg} &= 1 \text{ kg} \times \frac{c^2}{\text{MeV}} \times \frac{\text{MeV}}{c^2} \\ &= \frac{1 \times (3.0 \times 10^8)^2}{10^6 \times 1.6 \times 10^{-19}} \\ &= 5.625 \times 10^{29} \text{ MeV}/c^2 \end{aligned}$$

120.1 Convert 1 kg to MeV/c<sup>2</sup>.

--- ALTERNATIVE:

$$1 \text{ MeV}/c^2 = \frac{10^6 \times 1.6 \times 10^{-19}}{(3 \times 10^8)^2} = 1.778 \times 10^{-36} \text{ kg}$$

$$1 \text{ kg} = \frac{1}{1.778 \times 10^{-36}} = 5.625 \times 10^{29} \text{ MeV}/c^2$$

(no need to remember the final value, only the formula)

|       |                                                         |                                                                |                                               |                                               |
|-------|---------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|
|       |                                                         | I                                                              | II                                            | III                                           |
| 121.1 | What are the six quarks (two in each generation)?       | Up (u)<br>Down (d)                                             | Charm (c)<br>Strange (s)                      | Top (t)<br>Bottom/Beauty (b)                  |
| 122.1 | What are the six leptons (two in each generation)?      | Electron (e)<br>Electron Neutrino ( $\nu_e$ )                  | Muon ( $\mu$ )<br>Muon Neutrino ( $\nu_\mu$ ) | Tau ( $\tau$ )<br>Tau Neutrino ( $\nu_\tau$ ) |
| 123.1 | What are the four bosons (the "gauge"/"vector bosons")? | Photon ( $\gamma$ )<br>Gluon (g)<br>Z boson (Z)<br>W boson (W) |                                               |                                               |
| 124.1 | Symbol of the up quark                                  | u                                                              |                                               |                                               |
| 125.1 | Symbol of the down quark                                | d                                                              |                                               |                                               |
| 126.1 | Symbol of the electron                                  | e                                                              |                                               |                                               |
| 127.1 | Symbol of the electron neutrino                         | $\nu_e$                                                        |                                               |                                               |
| 128.1 | Symbol of the photon                                    | $\gamma$                                                       |                                               |                                               |
| 129.1 | What particles have a corresponding anti-particle?      | All of the fermions, and also the W boson                      |                                               |                                               |

|       |                                                                                            |                                                                                                                                                                                                                                             |
|-------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 130.1 | What are leptons?                                                                          | Fermions (spin-1/2 particles) that <u>do not</u> participate in the strong nuclear force.<br><i>That is, the electron and its generations, and the neutrino and its generations.</i>                                                        |
| 131.1 | What are quarks?                                                                           | Fermions that <u>do</u> participate in the strong nuclear force.<br><i>That is, the quarks.</i>                                                                                                                                             |
| 132.1 | What are fermions?                                                                         | Matter particles                                                                                                                                                                                                                            |
| 133.1 | What are bosons?                                                                           | Force carrying particles                                                                                                                                                                                                                    |
| 134.1 | What are hadrons?                                                                          | <u>Composite</u> particles (that is, not elementary particles) made of quarks.<br><i>Like, protons and neutrons are hadrons.</i>                                                                                                            |
| 135.1 | What is a baryon?                                                                          | A hadron made of a combination of <u>three</u> quarks.<br><i>Like, protons and neutrons</i>                                                                                                                                                 |
| 136.1 | What is a meson?                                                                           | A hadron made from a combination of a quark and an anti-quark.                                                                                                                                                                              |
| 137.1 | What is a pion?                                                                            | A meson made from any combination an up or down quark, and an anti-up or anti-down quark.                                                                                                                                                   |
| 138.1 | What is a kaon?                                                                            | A meson made from any combination of an up or down quark/anti-quark, and a strange or anti-strange quark.                                                                                                                                   |
| 139.1 | How are anti-particles symbolically notated?                                               | With a bar above the particle symbol (e.g. an anti-down quark is $\bar{d}$ )<br>...with the <b>sole exception</b> of the anti-electron, which is referred to as a " <u>positron</u> " and is notated as $e^+$                               |
| 140.1 | What quantities must be conserved for a particle interaction to be theoretically possible? | <ul style="list-style-type: none"> <li>• Charge</li> <li>• Baryon number</li> <li>• Lepton number</li> <li>• Strangeness (<i>though strangeness DOES NOT need to be conserved in WEAK interactions</i>).</li> </ul>                         |
| 141.1 | What is the Baryon number?                                                                 | A quantity used in conservation laws for particle interactions.<br>Every quark has a baryon number of $\frac{1}{3}$<br>Every anti-quark has a Baryon number of $-\frac{1}{3}$<br>All other fundamental particles have a Baryon number of 0. |
| 142.1 | What is the Lepton number?                                                                 | A quantity used in conservation laws for particle interactions.<br>Every lepton has a Lepton number of 1<br>Every anti-lepton has a Lepton number $-1$<br>All other fundamental particles have a Lepton number of 0.                        |
| 143.1 | What is strangeness?                                                                       | A quantity used in conservation laws in particle interactions.<br>The strange quark has a strangeness of $-1$ ( <u>NOT 1</u> )<br>The anti-strange quark has a Strangeness of 1<br>Every other fundamental particle has a Strangeness of 0. |

|       |                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 144.1 | What rule constrains the possible hadrons that can be formed by quarks?<br>(We're not considering color-charge theory)                                | <b>Hadrons must have an integer charge.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 145.1 | How can the following be checked for validity by conservation of charge?<br>${}_{92}^{235}\text{U} \Rightarrow {}_{90}^{231}\text{Th} + {}_2^4\alpha$ | The proton numbers (atomic number) also denote charge (since protons have a charge of 1).<br>$92 \equiv 90 + 2$ so this is valid.<br><br>For particles in these interaction equations that <i>aren't</i> atoms, we still use the bottom number, but have it mean charge instead of "proton count."                                                                                                                                                                                                                                                      |
| 146.1 | Verify the following by charge conservation, Baryon number conservation, and Lepton number conservation:<br>$d \Rightarrow u + e^- + \nu_e$           | <b>Charge conservation:</b> LHS: $-\frac{1}{3}$ RHS: $\frac{2}{3} + (-1) + 0 := -\frac{1}{3}$ is correct; charge is conserved.<br><br><b>Baryon number conservation:</b> LHS: $\frac{1}{3}$ RHS: $\frac{1}{3} + 0 + 0 := \frac{1}{3}$ is correct; Baryon number is conserved.<br><br><b>Lepton number conservation:</b> LHS: 0 RHS: $0 + 1 + 1 := 2$ is <i>incorrect</i> ; Lepton number is <b>not conserved</b> . That means this particle interaction <u>cannot happen</u> .                                                                          |
| 147.1 | What three physical quantities must be conserved in all particle interactions?                                                                        | <ul style="list-style-type: none"> <li>• momentum</li> <li>• mass-energy (<i>since mass and energy are equivalent</i>)</li> <li>• charge</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                     |
| 148.1 | What might affect the observed lifetime of a particle? ( <i>And how is this nicely convenient for us?</i> )                                           | When a particle travels at relativistic speeds, according to Einstein's theory of special relativity, time will move slower for it, from our perspective ( <i>if the particle had the ability to contemplate its situation, it would also see us as moving slower through time</i> ).<br><br>Exotic particles that decay quickly, will hence decay much slower than otherwise expected, when moving at relativistic speeds, making them easier for us to observe.<br>It also can allow many additional interactions with the exotic particle to happen. |
| 149.1 | What is kinetic theory? (3)                                                                                                                           | That matter (solids, gases, liquids) consist of a <i>very large number</i> of very small <b>particles</b> ( <i>referred to as molecules</i> )<br>...that are in <i>continuous</i> , <b>random motion</b><br><br>...and the idea that the <i>consideration of the microscopic movements</i> of these particles will <i>predict the macroscopic behaviour</i> of a substance.                                                                                                                                                                             |
| 150.1 | What is absolute zero?                                                                                                                                | The minimum possible temperature of a substance;<br><br>the point at which <i>all</i> of the molecules/[particles] have <i>no movement</i> ( <i>relative to you</i> )<br>OR the point at which all of the internal [kinetic] energy has been removed from the substance                                                                                                                                                                                                                                                                                 |
| 151.1 | What is "absolute temperature"?                                                                                                                       | A value of temperature on an <i>absolute temperature scale</i> ; a scale that has absolute zero with the value 0—that "starts at" absolute zero.                                                                                                                                                                                                                                                                                                                                                                                                        |
| 152.1 | Kelvin to °C                                                                                                                                          | $^{\circ}\text{C} = \text{K} - 273.15$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 153.1 | °C to Kelvin                                                                                                                                          | $\text{K} = ^{\circ}\text{C} + 273.15$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

154.1 How may we [accurately] *calibrate* a thermistor—determine the actual resistance at various temperatures?

Place the thermistor in a **beaker of water**

Use a **Bunsen burner** (*single open gas flame*) and **ice** for *temperature control*

Use a [mercury] **thermometer** (*classic tube thermomet*), and an **ohmmeter** across the thermistor for *measurement*.

Using the Bunsen burner and ice to control the temperature, we take various readings at different temperatures (recorded from the thermometer) for the resistance across the thermistor (recorded from the ohmmeter).

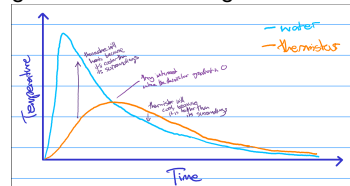
Due to **thermal inertia**, the temperature of the thermistor (*what the ohmmeter responds to*) may not be the same as the temperature of the water (*what the thermometer responds to*).

We can address this by

...stirring the water (*ensuring "homogeneity"*)

...keeping the thermometer close to the thermistor (*like, bundling them together*)

...recognising that the water and thermistor will be the same temperature when the ohmmeter reading (*a proxy for the thermistor temp.*) stops changing -> i.e. the inflexion point where it goes from increasing to decreasing or vice-versa;



$$R = R_0 e^{\frac{b}{T}}$$

Where:

155.1 Formula relating the resistance of a thermistor to its temperature

- $R_0$  and  $b$  are constants that need to be found/calibrated to the thermistor
- $R$  is the resistance of the thermistor
- $T$  is the absolute temperature of the thermistor (i.e. Kelvin)

Turning this into a linear equation we can plot:

$$\ln R = b \cdot \frac{1}{T} + \ln R_0$$

(i.e. we must plot  $\ln R$  on the y-axis and  $\frac{1}{T}$  on the x-axis)

156.1 Risk assessment for experiment calibrating a thermistor (CP12) (3x3)

**Hazard:** Hot water and fire, **Risk:** May cause flammable materials like clothing, hair and plastic coating on leads to catch fire/melt; hot water may cause burns, **Control:** Do not handle beaker when hot; tie up hair; always watch the experiment/ do not leave experiment unattended; keep leads away from hot things

**Hazard:** Glass apparatus, **Risk:** The beaker and/or thermometer can break if not handled carefully, **Control:** Keep beaker upright (don't let it roll); Keep it away from the edge of the bench

**Hazard:** Thermistor, **Risk:** By exceeding the voltage limit (*perhaps 6V*), the thermistor may be damaged, **Control:** Do not exceed 6V in potential divider circuit

157.1 What is heat?

Heat is the transfer of thermal energy. -> **it has a unit of energy.**

NOT that heat is an "energy" which objects have; 'heat energy'. Instead, that should be called *thermal energy*, or *internal energy*.

(*though in practice, people will do that anyway; referring to 'heat energy'*)

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | Transferring the same amount of heat<br>-> increases 'internal energy' by the same amount<br>-> increases temperature by <u>different</u> amounts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 158.1 | <p>How does a transfer of the same amount of heat [energy] to different materials affect their internal energy and temperature differently?</p> <p>temperature <math>\neq</math> internal energy (UNLESS IT'S AN IDEAL GAS)</p> <p><i>SO then, what is temperature? It describes the <u>average [translational] kinetic energy</u> of the molecules.</i></p> <p><i>Meanwhile, internal energy describes <u>all the ways molecules store energy</u>—'stretching of bonds' between molecules (intermolecular potential), rotational kinetic energy, vibrational kinetic energy, translational kinetic energy, etc.</i></p> <p><i>Adding internal energy to a system may use the energy in different sources, other than translational kinetic energy, and so there is a discrepancy between temperature and internal energy (except for ideal gases, where kinetic energy IS the only way for molecules to express energy)</i></p> |
| 159.1 | <p>What is 'specific heat capacity'? + Symbol + Unit</p> <p><b>'specific heat capacity':</b> A property of a substance; the energy required to raise the temperature of one kilogram of that substance by one degree kelvin/celsius.</p> <p><b>Symbol:</b> <math>c</math></p> <p><b>Unit:</b> <math>\text{J kg}^{-1}\text{K}^{-1}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|       | $\Delta E = mc\Delta\theta$ <p>Where:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 160.1 | <p>Specific heat capacity equation</p> <ul style="list-style-type: none"> <li>• <math>\Delta E</math> is the change in <i>internal energy</i> of the substance (J)</li> <li>• <math>m</math> is the mass of the substance (kg)</li> <li>• <math>\Delta\theta</math> is the change in temperature of the substance (<math>\text{K}/^{\circ}\text{C}</math>)</li> <li>• <math>c</math> is the <b>specific heat capacity</b> of the substance (<math>\text{J kg}^{-1}\text{K}^{-1}</math>)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 161.1 | <p>What does it mean that the specific heat capacity of a substance is higher?</p> <p>It means that it is <b>harder</b> to raise the temperature of that substance; it requires more energy to raise the temperature, for each degree.</p> <p><i>Hydrogen gas has the highest specific heat capacity (<math>14,300 \text{ J kg}^{-1}\text{K}^{-1}</math>), and that's because hydrogen molecules are so light, there's so many of them in each kilogram of hydrogen gas.</i></p> <p><i>Water is the next highest, followed by ice and steam and wood, and then air which is <math>1,020 \text{ J kg}^{-1}\text{K}^{-1}</math>. Metals follow, and have a MUCH LOWER specific heat capacity compared to water, for example. Meaning they are easier to heat up in terms of energy input.</i></p>                                                                                                                                  |
| 162.1 | <p>What is 'specific latent heat'? + Symbol + Unit</p> <p><b>'specific latent heat':</b> A property of a given phase transition; the energy required to change the <u>state</u> of one kilogram of a given substance to another given substance (or vice-versa), at a <i>constant temperature</i>.</p> <p><b>Symbol:</b> <math>L</math></p> <p><b>Unit:</b> <math>\text{J kg}^{-1}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 163.1 | <p>What is 'specific latent heat of fusion'?</p> <p>Refers to the specific latent heat of a phase transition that goes to/from a <i>solid</i> and a <i>liquid</i>.</p> <p><i>Useful when you want to talk about the specific latent heat of a specific LIQUID, going to/from a SOLID (i.e. disambiguating whether from the liquid you're talking about a phase transition to/from a solid, or to/from a gas). e.g. "the specific latent heat of fusion of water is <math>334 \text{ kJ/kg}</math>"</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                       |

Refers to the specific latent heat of a phase transition that goes to/from a *gas* and a *liquid*.

164.1 What is 'specific latent heat of vaporisation'?

*Useful when you want to talk about the specific latent heat of a specific LIQUID, going to/from a GAS (i.e. disambiguating whether from the liquid you're talking about a phase transition to/from a gas, or to/from a solid). e.g. "the specific latent heat of vaporisation of water is 2.26 MJ/kg"*

$$\Delta E = Lm$$

Where:

165.1 The specific latent heat equation

- $\Delta E$  is the amount of energy required to fully complete the phase change (for a substance at the *critical temperature*— so that no energy is required for extra heating/cooling)
- $m$  is the mass of the substance which we want to undergo phase transition
- $L$  is the **specific latent heat** of the substance, for the phase transition we are considering

The additional [...] input into a substance for a phase change

166.1 ...[...] increase the [...] of the substance ...for a [...]

Cover Previous  
Reveal Next

The additional **internal energy** input into a substance for a phase change

...**does not** increase the **temperature** of the substance

...for a '**pure substance**'

*(which is to say that other stuff mixed in with a different melting/boiling point is gonna keep changing temp. through the energy input)*

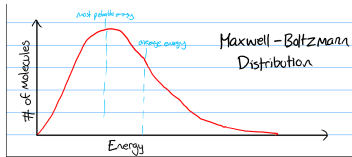
*(also, we are describing 'additional energy' input, but this is also true for phase changes where we REMOVE internal energy like for gas -> liquid, or liquid -> solid)*

167.1 Experimental method to determine the specific latent heat of fusion of water (CP13) (5)

1. **Crush** some ice, then place in a **funnel** to allow it to *warm up to 0°C*— **catching melted ice** in a *container*
2. Get a **dry, empty beaker** and a **mass balance** with a **thermometer** (for making measurements)
3. Add **water** to the beaker *first*, measuring the **mass of the water** added (*using the balance*), and measuring the **temperature of the water** (*using the thermometer*).
4. Then add the **crushed ice** *second*, measuring the **mass of the ice** added (*using the balance*).  
*Specifically, we'll have three mass values; mass of the dry, empty beaker, mass of the beaker + water, and mass of the beaker + water + ice. We can use these to find the mass of the water and [crushed] ice individually.*
5. **Record the coldest temperature seen on the thermometer.**

This will occur when the last of the ice melts.

*The thermal energy of the water is constantly being transferred to the ice for the phase change. Thus, the water temperature will continue to decrease until there's no more phase change to engage in— all the ice has melted. At thus point the water + melted ice temp. will start increasing (assuming the room temp. is warmer than it, which is probably true).*

|       |                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 168.1 | <p>Calculation for specific latent heat of fusion for water from <math>m_{\text{water}}</math>, <math>m_{\text{ice}}</math>, specific heat capacity of water, <math>c</math>, temperature of water before [crushed] ice is added, <math>\theta_1</math>, and the temperature of the water the moment the last of the ice has melted, <math>\theta_2</math>. (CP13)</p> | <p>heat lost by cooling water = heat gained by ice + warming melted ice</p> $\Delta E_{\text{cooling water}} = \Delta E_{\text{melting ice}} + \Delta E_{\text{warming new water}}$ <p>(we assume there's no <math>\Delta E</math> for "warming ice"—that is, that the ice is perfectly <math>0^\circ\text{C}</math>)</p> $m_{\text{water}} \times c \times \Delta\theta_{\text{before} \rightarrow \text{current}} = m_{\text{ice}} \times L + m_{\text{ice}} \times c \times \Delta\theta_{0 \rightarrow \text{current}}$ $m_{\text{water}} \times c \times (\theta_1 - \theta_2) = m_{\text{ice}} \times L + m_{\text{ice}} \times c \times (\theta_2 - 0)$ <p>(the LHS should have <math>(\theta_2 - \theta_1)</math> instead, but that would make it a negative change in energy—i.e. a loss of internal energy of the water, which is what it is! but we want it to equal the RHS; representing a <u>gain</u> in internal energy)</p> |
| 169.1 | <p>Safety considerations in finding the latent heat of fusion of water (CP13) (3)</p>                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• Clamp the thermometer carefully so that it cannot tip the beaker over</li> <li>• Do not use the thermometer as a stirrer.</li> <li>• Clear up any spilt ice or water on the floor to avoid slipping</li> </ul> <p>TODO: This card can be improved</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 170.1 | <p>What is internal energy?</p>                                                                                                                                                                                                                                                                                                                                        | <p>The sum of <i>all</i> kinetic <i>and</i> potential energies of all <u>molecules</u> within a given mass of substance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 171.1 | <p>What is the Maxwell-Boltzmann distribution? What does the distribution tell us?</p>                                                                                                                                                                                                                                                                                 | <p>The distribution of energies of particles/molecules in an <u>ideal gas</u>, at a <b>given temperature</b>.</p>  <p>(a graph of "no. molecules" on the y-axis, and "energy" on the x-axis. The distribution passes through <math>(0,0)</math>, and resembles a bell curve, though has a tail off to the right).</p> <ul style="list-style-type: none"> <li>• There are no molecules with zero energy</li> <li>• Only a few molecules have high energy</li> <li>• There is no maximum value for the energy a molecule can have</li> </ul>                                                                                                                                                                                                                                                                                                               |

- 172.1 Explain the equation  

$$P = \frac{1}{3} \cdot \frac{N}{V} m \langle u^2 \rangle$$
- $\langle u^2 \rangle$  is the *average squared speed* of molecules. Hence, **pressure is proportional to velocity squared**. Because there are two contributing factors:
    - molecules can move farther in a given length of time, increasing the number of collisions with the wall [per unit time] -> increasing pressure
    - increasing the velocity increases the force exerted in each collision -> increasing pressure
  - **Pressure is proportional to the mass of molecules,  $m$ .** Because, increasing mass increases *impulse* behind each collision with a wall -> increasing pressure
  - **Pressure is proportional to the number of molecules,  $N$ .** Because, increasing the number of molecules (within a given volume) means more collisions with the wall [per unit time] -> increasing pressure.
  - **Pressure is inversely proportional to volume [that the gas occupies],  $V$ .** Because, decreasing volume increases the density of molecules (given a number of molecules), leading to more collisions with the wall per second -> increasing pressure.
  - The  $\frac{1}{3}$  factor comes from **space being 3-dimensional**. Meaning, that a *third* of the mean-squared speed of a molecule (which will be moving in a *random direction*), on average, will contribute to the pressure in any given direction. Mathematically,  $u^2 = u_x^2 + u_y^2 + u_z^2$ ;  $\langle u^2 \rangle = \langle u_x^2 \rangle + \langle u_y^2 \rangle + \langle u_z^2 \rangle = 3\langle u_x^2 \rangle$  (since the average velocity will be the same in any direction);  $\langle u_x^2 \rangle = \frac{1}{3} \langle u^2 \rangle$ .

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$$

Where:

- 173.1 Equation relating average kinetic energy of molecules with the temperature of an ideal gas. + Derivation
- $\frac{1}{2} m \langle c^2 \rangle$  is the mean kinetic energy in the gas.
    - $m$  is the mass of each molecule
    - $\langle c^2 \rangle$  is the average squared speed of the molecules
  - $k$  is the Boltzmann constant ( $1.38 \times 10^{-23} \text{ JK}^{-1}$ )
  - $T$  is the **absolute temperature** of the gas (i.e. it *must be in Kelvin*)

Derivation:

Combine  $P = \frac{1}{3} \cdot \frac{N}{V} m \langle u^2 \rangle$  and  $PV = NkT$ :

$$\frac{1}{3} Nm \langle u^2 \rangle = NkT$$

$$m \langle u^2 \rangle = 3kT$$

$$\frac{1}{2} m \langle u^2 \rangle = \frac{3}{2} kT$$

- 174.1 The temperature of a gas is proportional to the molecules' [...].

The temperature of a gas is proportional to the molecules' **average kinetic energy**.

- 175.1 What is root-mean-square speed? + abbreviation + symbolic notation

root-mean-square speed (**rms**) is a measure of the *average speed* of gas molecules. It is notated as  $\sqrt{\langle c^2 \rangle}$  ...with the angle brackets indicating to take an arithmetic mean.

- 176.1 What is Boyle's Law?

The **pressure** exerted by a gas is *inversely proportional* to the **volume** it occupies ...for a *constant mass*, at a constant temperature.

$$p \propto \frac{1}{V}$$

The **volume** occupied by a gas is *proportional to* its **absolute temperature**  
...for a *constant mass*, at a constant pressure.

177.1 What is Charles' Law?

$$V \propto T$$

a.k.a. the **Pressure Law**

The **pressure** exerted by a gas is *proportional to* its **absolute temperature**  
...for a *constant mass*, at a constant volume.

178.1 What is Gay-Lussac's Law? (+alternative name)

$$p \propto T$$

179.1 What are some of the requirements for an 'ideal gas'? (5/7)

1. The molecules have negligible size
2. The molecules are all identical
3. All collisions are perfectly elastic
4. The time of a collision is significantly smaller than the time between collisions
5. The molecules exert no forces on each other, except during collisions
6. There are enough molecules so that statistics can be applied
7. The motion of the molecules is random

180.1 What is the Boltzmann constant?

$$k = 1.38 \times 10^{-23} \text{ JK}^{-1}$$

181.1 What is the specific heat capacity of air (in typical room conditions)?

$$c = 1,012 \text{ J kg}^{-1}\text{K}^{-1}$$

$$pV = NkT$$

Where:

182.1 What is the 'equation of state for ideal gases'?

- $p$  is the pressure exerted by the gas ( $\text{Pa}$ ;  $\text{Nm}^{-2}$ )
- $V$  is the volume the gas occupies ( $\text{m}^3$ )
- $N$  is the number of molecules in the gas (unitless count)
- $k$  is Boltzmann's constant ( $1.38 \times 10^{-23} \text{ JK}^{-1}$ )
- $T$  is the **absolute temperature** of the gas ( $\text{K}$ )

$$pV = nRT$$

183.1 What is the equation of state for ideal gases in terms of moles?

Where:

- $p$  is the pressure exerted by the gas ( $\text{Pa}$ ;  $\text{Nm}^{-2}$ )
- $V$  is the volume the gas occupies ( $\text{m}^3$ )
- $n$  is the number of *moles* of the gas (unitless count)
- $R$  is a constant ( $8.3076 \text{ JK}^{-1}$ )
- $T$  is the **absolute temperature** of the gas ( $\text{K}$ )

A more practical SI unit measuring the "amount of" a substance, abbreviated to mol. 1 mole is defined to be an "Avogadro's number" number of molecules of that substance:

184.1 What is a mole?

$$1 \text{ mol} = 6.02 \times 10^{23} \text{ molecules}$$

185.1 What is meant by an ideal gas?

A gas that obeys the equation  $PV = nRT$  /  $PV = NkT$

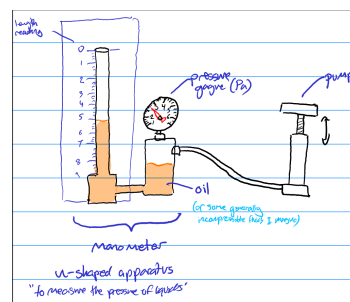
*Saying it is a "gas which perfectly follows the gas laws" isn't quite strong enough.*

186.1 Experimental method for (accurately) demonstrating Boyle's law (CP14)

1. Use a **manometer** filled with **oil** (or another generally incompressible fluid), trapping *two separate volumes of air*, on either side (-> *neither side is open to atmosphere*). On one side the tube is transparent and has **length readings**. On the other we are able to attach a **pressure gauge** and a **pump**.
2. Pump a bit of air into the manometer at a time. **Wait some time** (to let the temperature settle back down to room temperature), and take a reading of the [gauge] **pressure** and the **length of air** on the other side.

*As we pump air into the right side, we're depositing an excess of air molecules into a confined volume— this increases the pressure exerted by the air gas onto its surroundings, including the oil lying beneath it; the oil is pushed down and forced up the left-hand side tube. There, the oil meets another air pocket, which it will compress in turn until the pressure exerted by it and the pressure exerted by the right-side pocket balance. This means the pressure read by the gauge and the pressure of the air pocket we're taking a length reading for are equivalent.*

3. To not deal with *liquid getting stuck on the side of the tube*, we **only ever increase the pressure**. And we **don't take repeat measurements**. To compensate, we **take more than the usual 6-8 measurements**.
4. By plotting **absolute pressure** vs. **1 / length** we should observe a straight line passing through the origin, confirming Boyle's Law that pressure and volume are inversely proportional.  
*We can use length as a proxy for volume because the radius of the tube is constant, And from  $V = \text{length} \cdot \pi r^2$ , that means volume is proportional to length.*
5. We also mitigate **parallax errors** by reading the column of liquid at eye level, and we mitigate **meniscus errors** by reading from the bottom of the meniscus. We recognise that a *non-0 y-intercept* in our plot represents a **systematic error** (perhaps from the length reading, or the pressure gauge, or other).



(presumably this picture shows some air already having been pumped in— as otherwise we expect the oil level on both sides to equalize)

|       |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 187.1 | Risk assessment for experiment demonstrating Boyle's Law (CP14) (2x3)                               | <p><b>Hazard:</b> Apparatus could fall over (especially the tall tube with high centre of gravity), <b>Risk:</b> Injury to feet, <b>Control:</b> Clamp apparatus to the bunch; keep away from edge</p> <p><b>Hazard:</b> Tubing and/or joints could burst (especially since we're dealing with high pressure, <b>Risk:</b> Tubing and/or parts striking eyes, <b>Control:</b> Observe apparatus carefully during experiment; wear safety goggles.</p>                                                                                                                                                                                                                                                                                        |
| 188.1 | Explain how changes to the volume of your chest cavity are able to draw air in, and breathe air out | <p>Increasing volume decreases <b>pressure</b>. -&gt; The difference from the atmospheric pressure causes an influx of air, in order to equalise the pressure.</p> <p>Decreasing volume by contracting chest cavity increases <b>pressure</b>. -&gt; The difference from the atmospheric pressure causes an outflow of air, in order to equalise the pressure.</p> <p><i>We may also explain that in <math>pV = NkT</math>, the pressure is constant, the temperature is constant, hence <math>V \propto N</math> and when the volume increases the amount of air (gas) in the chest cavity must increase. And when the volume decreases the amount of air in the chest cavity must decrease; air must leave i.e. you "breathe out."</i></p> |